

Zhentao Liu

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Bridging Advanced 3D/4D Computer Vision with Medical Image Reconstruction

Education

ShanghaiTech University — Shanghai, China
Ph.D. in Computer Science - GPA: 3.62/4 — 2022.09 - Present

- IMPACT Lab | Supervisor: Prof. Dinggang Shen, Prof. Zhiming Cui
- Research Interests: 3D/4D Reconstruction, NeRF/3DGS/Feed-Forward in Medical Image Reconstruction

Ningbo University — Ningbo, China
B.E. in Communication Engineering - GPA: 3.83/4, Rank 1/100 — 2018.09 - 2022.06

- VIPP Lab | Supervisor: Prof. Qiuping Jiang
- Research Interests: Human Vision Modeling, Image Quality Assessment

Selected Projects

Sparse-View Dynamic DSA Reconstruction — 2023.05 - Present

- **Task Definition**: Reconstruct high-fidelity 3D vessels from sparse dynamic Digital Subtraction Angiography (DSA) images (**133** → **30 views**) to reduce radiation exposure and support medical assessment.
- **VPAL** *MIA Under Review* Paper 2023.05 – 2024.03
  A **NeRF-based** framework that introduces a **vessel probability neural field** to describe vessel occupancy. This innovation ensures accurate gradient flow and enables static-dynamic decomposition, significantly improving reconstruction quality.
- **4DRGS** *IPMI 2025 Oral* Paper · Code · Page 2024.07 – 2024.12
  The first 3DGS-based framework for efficient 3D vessel reconstruction. Develop **4D radiative Gaussian kernels** and a novel pruning strategy for enhanced reconstruction quality. Achieve SOTA results within **minutes**, **>10x speedup** compared to VPAL.
- **Feed-Forward 3DGS** *Ongoing Project* 2025.11 – Present
  Aim to develop a generalizable feed-forward 3DGS model to accelerate the reconstruction process to **seconds**, enabling real-time clinical applications.
- **Future Outlook**: Establish a **Vessel World Model** for **autonomous endovascular navigation** via Deep Reinforcement Learning, aiming to automate guidewire control for precise surgeries and **Interventional BCI** implantation.

Cardiac CT Motion Artifact Reduction — 2024.12 – Present

- **Task Definition**: Reduce cardiac CT motion artifact from short-scan data to support medical assessment.
- **GDVox** *IEEE TMI In-Preparation*
  The first neural rendering-based framework to represent cardiac motion. It decomposes the dynamic heart into a static canonical voxel grid driven by a sparse set of learnable Gaussian deformation kernels.

Sparse-view CBCT Reconstruction — 2022.09 – 2023.03

- **Task Definition**: Reconstruct high-fidelity 3D CBCT image from sparse X-ray images (**500** → **20 views**) to reduce radiation exposure.
- **Geometry-Aware Reconstruction** *IEEE TMI 2024* Paper · Code & Dataset · Page
  Inspired by NeRF, propose a geometry-aware encoder-decoder framework that reconstructs 3D CBCT from sparse X-rays without per-scene optimization. By enforcing **multi-view epipolar geometry** during voxel feature querying, the model enables high-quality reconstruction **within one second** via feed-forward inference.

Selected Publications

4DRGS: 4D Radiative Gaussian Splatting for Efficient 3D Vessel Reconstruction from Sparse-View Dynamic DSA Images
Zhentao Liu*, Ruyi Zha*, Huangxuan Zhao, Hongdong Li, Zhiming Cui.
Information Processing in Medical Imaging (**IPMI, Oral**), 2025.

Geometry-Aware Attenuation Learning for Sparse-View CBCT Reconstruction
Zhentao Liu, Yu Fang, Changjian Li, Han Wu, Yuan Liu, Dinggang Shen, Zhiming Cui.
IEEE Transactions on Medical Imaging (**IEEE TMI**), 2024.

3D Vessel Reconstruction from Sparse-View Dynamic DSA Images via Vessel Probability Guided Attenuation Learning
Zhentao Liu, Huangxuan Zhao, Wenhui Qin, Zhenghong Zhou, Xinggang Wang, Wenping Wang, Xiaochun Lai, Chuansheng Zheng, Dinggang Shen, Zhiming Cui.
Medical Image Analysis (**MIA**), Under Review.

3D MedDiffusion: A 3D Medical Latent Diffusion Model for Controllable and High-quality Medical Image Generation
Haoshen Wang, **Zhentao Liu**, Xiaodong Wang, Kaicong Sun, Dinggang Shen, Zhiming Cui.
IEEE Transactions on Medical Imaging (**IEEE TMI**), 2025.

Structure-Preserving Two-Stage Diffusion Model for CBCT Metal Artifact Reduction
Xingyue Wang, **Zhentao Liu**, Haoshen Wang, Minhui Tan, Zhiming Cui.
IEEE Transactions on Medical Imaging (**IEEE TMI**), 2025.

Towards Top-Down Just Noticeable Difference Estimation of Natural Images
Qiuping Jiang*, **Zhentao Liu***, Shiqi Wang, Feng Shao, Weisi Lin.
IEEE Transactions on Image Processing (**IEEE TIP**), 2022.

Single Image Super-Resolution Quality Assessment: A Real-World Dataset, Subjective Studies, and An Objective Metric
Qiuping Jiang, **Zhentao Liu**†, Ke Gu, Feng Shao, Xinfeng Zhang, Hantao Liu, Weisi Lin.
IEEE Transactions on Image Processing (**IEEE TIP**), 2022. (**TOP50 Popular Articles**)

Mask2Surface: Motion Correction and Super-Resolution for Cardiac Surface Reconstruction Using Latent Diffusion
Zichen Zhang, **Zhentao Liu**, Zeng Zhang, Zhiming Cui.
Medical Image Computing and Computer Assisted Intervention (**MICCAI, Early Accept**), 2025.

TeethDreamer: 3D Teeth Reconstruction from Five Intra-oral Photographs
Chenfan Xu*, **Zhentao Liu***, Yuan Liu, Yulong Dou, Jiamin Wu, Jiepeng Wang, Minjiao Wang, Dinggang Shen, Zhiming Cui.
Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024.

Honors & Awards

Ningbo Natural Science Outstanding Paper Award (1st Prize) 2023
Selected as the best paper out of 130 candidates (1/130). Ningbo, China

Zhejiang Provincial Government Scholarship 2019, 2020
Awarded to top 3% students based on academic excellence. Zhejiang, China

First Class Academic Scholarship 2021
Awarded to top 5% students in Ningbo University. Ningbo, China

Reviewer Services

- IEEE Transactions on Medical Imaging (**IEEE TMI**)
- IEEE Journal of Biomedical and Health Informatics (**IEEE JBHI**)
- Medical Image Computing and Computer Assisted Intervention (**MICCAI**)
- International Workshop on Multimedia Signal Processing (**MMSP**)
- CVPR MetaFood Workshop
- International Conference on Machine Learning in Medical Imaging (**MLMI**)

Technical Skills

- **Programming**: Python, C/C++, CUDA, MATLAB
- **Research Tools**: Pytorch, Meshlab, 3D Slicer, ITK-SNAP, LaTeX
- **Language**: English (fluent), Chinese (native)